

Reetesh Mukul

Large Multimodal Models | Document AI | Computer Vision

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Profile

Senior Principal Applied Scientist building production AI systems across large multimodal models, document AI, computer vision, and high-performance software. Leads applied research from problem formulation and model architecture through training, evaluation, and deployment, with current work spanning visual grounding, video reasoning, deepfake detection, and LLM-powered policy verification.

Core Expertise

Large Multimodal Models. Vision-language adaptation, visual grounding, cross-modal alignment, temporal reasoning, and evaluation design.

AI Guardrails and Policy Verification. LLM-powered policy and query representations, query-to-policy mapping, constraint-aware reasoning, and multi-step action evaluation.

Document Intelligence. Schema-guided extraction, multilingual OCR, complex layouts, reading order, tables, and constrained key-value extraction.

Scientific and Visual Learning. Bayesian uncertainty estimation, causal modeling, reinforcement learning, synthetic-media detection, face anti-spoofing, and adaptive routing.

Research Highlights

Policy verification for AI systems. Develops an LLM-powered policy verifier for AI guardrails, using structured policy and query representations, query-to-policy mapping, constraint-aware reasoning, and multi-step action evaluation.

Structured document intelligence. Develops schema-guided LMM workflows for multilingual OCR, complex layouts, reading order, tables, and constrained key-value extraction.

Visual authenticity and video reasoning. Studies multimodal evidence for deepfake detection, face anti-spoofing, and frame-level video analysis, with an emphasis on robust evaluation.

Adaptive visual systems. Combines vision transformers, Bayesian learning, diffusion methods, reinforcement learning, and routing to model memorability, improve creative content, and optimize system behavior.

Professional Experience

Senior Principal Applied Scientist

Dec 2023-Present

Oracle India | Bengaluru

Lead applied research for large multimodal models, from problem formulation and model design through training, evaluation, and productization. Current work includes LLM-powered policy verification for AI guardrails, schema-guided document intelligence for multilingual OCR, invoices, and tables, plus biometric deepfake detection and frame-level video reasoning.

Principal Machine Learning Engineer

Dec 2022-Nov 2023

Huawei Singapore Research Center | Singapore

Led research-to-prototype work on advertising creative improvement using diffusion-based inpainting and outpainting. Developed visual-content understanding and ad-effectiveness models grounded in attention, saliency, memorability, Bayesian reasoning, and vision transformers.

Principal Machine Learning Engineer

May 2021-Nov 2022

Huawei India Research Center | India

Led research on image memorability and cognitive aspects of visual content, connecting advertising-effectiveness signals to click-through-rate prediction. Applied Bayesian probabilistic programming and trained dense vision-transformer models for image memorability.

Senior Computer Scientist

Sep 2016-Apr 2021

Adobe | India

Led applied ML and performance research for creative applications, including visual similarity, panoptic segmentation, and interactive systems. Developed reinforcement-learning and multi-armed-bandit methods for adaptive scheduling, resource management, and job-executor control; co-invented techniques for responsive input, UI scheduling, teeth-region detection, and application-performance management.

Senior Software Developer / ML Engineer

2015-2016

Flipkart | India

Developed ad click-prediction models for CTR, CVR, ranking, and sampling, and contributed to the Android native-ad SDK.

Software Developer to Senior Software Developer

2008-2015

Qualcomm | India

Built OCR for Indic Devanagari characters, scene-detection capabilities for camera workflows, voice state-machine components, and embedded drivers.

Software Developer

2006-2008

Texas Instruments | India

Developed software for GPS components in embedded environments.

Selected Patents and Publications

Schema-guided LMM key-value extraction Oracle | US 2026/0127165 A1 | 2026 | Vision-transformer image memorability Oracle | US 2026/0105736 A1 | 2026 | Multimodal video highlight reasoning Oracle | US 2026/0187146 A1 | 2026 | RL-based resource management Adobe | US 11,556,393 B2 | 2023 | Bandit-controlled job-executor policies Adobe | US 12,547,446 B2 | 2026 | WiMAX QoS bandwidth requests IFIP WOCN | 2006

Education

M.Tech, IIT Bangalore | B.Tech, BIT Sindri